ISPhO

Thermal oxidation

Part A. Theoretical principles of ellipsometry (4.5 points)

A.1 (0.2pt)

A.2 (0.4pt)

$k_{1x} =$

$k_{2x} =$

$k_{3x} =$

A.3 (0.6pt)

Equation:

A.4 (0.6pt)

$r_s =$

$A =$ $B =$

A.5 (0.6pt)

Equation:

A.6 (0.6pt)

$r_p =$

$A =$ $B =$

A.7 (1.5pt)

$\theta_{\text{opt}} =$

Part B. Kinetics of oxidation (5.5 points)

B.1 (0.7pt)

$\Phi =$ $d =$

B.2 (0.7pt)

$\Phi =$ $d =$

B.3 (0.3pt)

t, min	d, nm

B.4 (0.3pt)

$v =$

B.5 (0.5pt)

$C_0 =$

$C_i =$

B.6 (0.5pt)

Equation:

$A =$ $B =$

$\tau =$

B.7 (1.0pt)

T	B	B/A

B.8 (1.5pt)

$\Delta E_b =$ $\Delta E_a =$

Plot:

B

B/A

Theory

ISPhO

A2-1

English

Memristors and NVRAM

Part A. Memristor

A.1

$\frac{dx}{dt} =$

A.2

$x_{0,\min} =$

A.3

$\tau =$

A.4

$Q =$

Part B. Memristor hysteresis

B.1

$x(t) =$

$\Delta I(\varphi) =$

B.2

$s =$

B.3

$\alpha =$ $\xi =$

Part C. ReRAM energy efficiency

C.1

$\tau =$

C.2

$Q =$

Part D. Segnetoelectrics and FeRAM

D.1

$E_{\mathrm{cr}} =$ $P_{\mathrm{cr}} =$

D.2

$P_{\mathrm{after}} =$

D.3

$Q =$

D.4

(Formula) $Q_{1\,\mathrm{TiB}} =$ (Value) $Q_{1\,\mathrm{TiB}} =$

Theory

ISPhO

A3-1

English

Hydraulic jump

Part A. Flow without energy loss

A.1

$E =$

A.2

$d_c =$

A.3

$v_c =$ $\mathrm{Fr}_c =$

A.4

Equation:

A.5

$\Delta v/v_1 =$ $\Delta d/d_1 =$

A.6

Velocity increases for:

Velocity decreases for:

Part B. Theory of hydraulic jumping

B.1

$F =$

B.2

Equation:

B.3

$d_2/d_1 =$

B.4

$\Delta E =$

B.5

A jump is possible for:

Part C. Hydraulic jump in the sink

C.1

(Formula) $v =$ (Value) $v =$

C.2

$d(r) =$

$d(r_c) =$

C.3

$\mathrm{Fr} =$ $d_2/d_1 =$